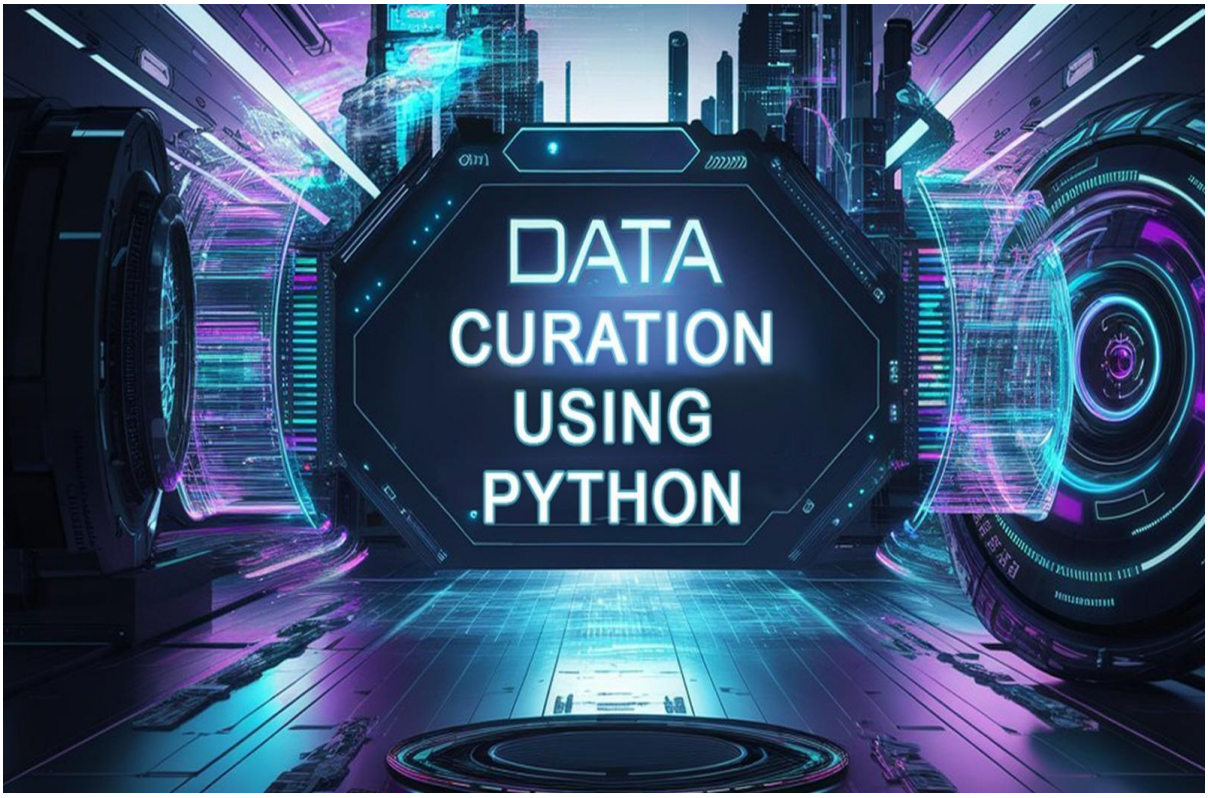

Model Curriculum

Fundamentals of Data Curation using Python

QP Code: NIE/ITS/N1016

Version: 1.0

NSQF Level: 3.5

Training Parameters

<table>
<tr><td>Sector</td><td>Information Technology Sector</td></tr>
<tr><td>Sub-Sector</td><td>Data Science & Engineering</td></tr>
<tr><td>Occupation</td><td>Data Curator Assistant</td></tr>
<tr><td>Country</td><td>India</td></tr>
<tr><td>NSQF Level</td><td>3.5</td></tr>
<tr><td>Aligned to NCO/ISCO/ISIC Code</td><td>NCO/2015-2512.0205</td></tr>
<tr><td>Minimum Educational Qualification and Experience</td><td>
<table>
<tr><th>Academic/Skill Qualification (with Specialization - if applicable)</th><th>Required Experience (with Specialization - if applicable)</th></tr>
<tr><td>Grade 10 pass and pursuing continuous schooling*</td><td>No experience required</td></tr>
<tr><td>Grade 8 with 2 years of NTC in the field of IT Sector*</td><td>No experience required</td></tr>
<tr><td>Grade 8th pass*</td><td>3 Year relevant experience in IT</td></tr>
<tr><td>Previous NSQF Level 3 Required</td><td>1.5-year relevant experience in IT</td></tr>
</table>
Prerequisite: Fundamentals of Python
*Should have a basic understanding of coding.
</td></tr>
<tr><td>Pre-Requisite License or Training</td><td>NA</td></tr>
<tr><td>Minimum Job Entry Age</td><td>16 years</td></tr>
<tr><td>Last Reviewed On</td><td>18.02.2025</td></tr>
<tr><td>Next Review Date</td><td>18.02.2028</td></tr>
<tr><td>NSQC Approval Date</td><td>18.02.2025</td></tr>
<tr><td>SNOS Version</td><td>1.0</td></tr>
<tr><td>Model Curriculum Creation Date</td><td>18.02.2025</td></tr>
<tr><td>Model Curriculum Valid Up to Date</td><td>18.02.2028</td></tr>
<tr><td>Model Curriculum Version</td><td>1.0</td></tr>
<tr><td>Duration of the Course</td><td>120 hours</td></tr>
</table>

Overview

This section summarises the end objectives of the program along with its duration

Training Outcomes of the course:

Students will gain proficiency in Python programming and foundational Artificial Intelligence concepts, including their applications and ethical considerations. They will master data curation fundamentals, encompassing collection, cleaning, preprocessing, integration, and storage using advanced tools and techniques. The course emphasizes ensuring data quality and governance, along with effective data visualization and reporting skills using tools like Tableau and Power BI. Participants will apply their knowledge through real-world projects and develop career readiness with insights from industry experts.

Compulsory Modules

The table lists the modules and their duration corresponding to the Compulsory NOS of the Qualification file.

NOS/Module Name	Training Duration (Hours)		
	Theory	Practical	Total
NOS-1/Module-1: Fundamentals of Data Curation using Python NOS Code: NIE/ITS/N1016 NSQF Level: 3.5	45	75	120
Total	**45**	**75**	**120**

Module Details

Module1/NOS1: Fundamentals of Data Curation using Python

NOS Code: NIE/ITS/N1016

Terminal Outcomes:

After completion of the module, the students shall be

- Able to acquire proficiency in Python programming for real-world data projects and understand AI fundamentals, applications, and ethical considerations.
- Comprehend data curation principles, data types, and related tools, learn various ethical methods for data collection and acquisition and perform data cleaning, handling missing data, and applying transformation techniques.
- Learn to Integrate and manage data storage using modern tools like DBMS and cloud solutions, ensure data quality through governance frameworks and quality metrics.
- Create effective visualizations and reports using tools like Tableau and Power BI , apply skills to real-world projects, collaborating and presenting findings and this will prepare them for careers in data curation with insights into industry trends and professional development.

Detailed Syllabus:

Name of the Topics	Theory (Hours)	Practical (Hours)	Total (Hours)
Section-1: Foundational in Python Programming, Artificial Intelligence & Data Science			
Foundational in Python Programming	10	20	30
Basics of Artificial Intelligence & Data Science	7	8	15
Section-2: Foundational in Data Curation			
Introduction to Data Curation	3	5	8
Data Collection & Acquisition Methods	13	10	23
Data Integration, Storage and Visualization	5	10	15
Data Quality and Governance	4	5	9
Advanced-Data Management Techniques	3	5	8
Application of Data Curation	0	12	12
Total Duration (in Hrs.)	**45**	**75**	**120**

Section-1: Foundational in Python Programming, Artificial Intelligence & Data Science

Foundational in Python Programming: 30 hours

- **Theory (10 Hours)**

  - Understanding to set up the Python environment.
  - Learning about variables and different data types in Python.
  - Understanding conditional statements (if, else, elif) and learning about loops (for, while).
  - Understanding the concepts of functions and learning to define and call functions.
  - Understanding the basics of file handling in Python.
  - Understanding python lists (creation, indexing, slicing, etc.) and dictionaries (creation, accessing, etc.), their properties, and uses.
  - Understanding the basics of Python libraries -NumPy, Matplotlib, and Pandas.

  Practical (20 Hours)

  - Learning to install Python and an IDE.
  - Understanding to write and execute simple python scripts such as "Hello, World!"
  - Learning to perform basic operations with variables and data types.
  - Learning to write programs using conditional statements and loops.
  - Learning to create and use functions in Python programs.
  - Learning to read from and write to files using Python.
  - Learning to perform operations with lists and dictionaries in Python.
  - Learning to perform basic operations using various Python libraries.

- Developing and implementing a project, integrating various concepts learned.

Basics of Artificial Intelligence & Data Science: 15 hours

- **Theory (07 Hours)**
  - Introduction to Artificial Intelligence
    - Understanding the basic concepts and evolution of Artificial Intelligence
    - Understanding the key components of Artificial Intelligence- Machine Learning, Deep Learning, Computer Vision & Natural Language Processing NLP.
    - AI in Healthcare, Finance, Education & Entertainment- Use cases & applications
    - Introduction to Generative AI – tools and use cases.
  - Basic Data Science and Statistics
    - Introduction to Data Science Concepts
    - Learning to handle large & complex datasets
    - Learning to implement data preprocessing techniques- Cleaning, Normalization, Handling Missing Data on sample datasets.
    - Understanding statistical Concepts: Distributions, Variance, Data Sampling
  - Ethical Considerations in AI
    - Introduction to AI Ethics and Biasness in AI models
    - Understanding Privacy Concerns with the use of AI
    - Understanding safe and responsible use of AI
- **Practical (08 Hours)**
    - Demonstration of popular AI tools
    - Hand on experience on open-source GenAI tools.
    - Demonstration of AI use cases in Healthcare, Finance, Education & Entertainment.
    - Introduction to low-code/no-code AI platforms
    - Various test cases to use preprocessing techniques- data cleaning, normalization, and handling missing data.

Section-2: Foundation in Data Curation

Introduction to Data Curation: 8 hours

- **Theory (3 Hours)**
  - Defining Data Curation and understanding the scope of data curation.
  - Evaluating the business needs and understanding the context for which the data is being curated.
  - Case studies showcasing the impact of effective data curation.
  - Learning about various types of Data (Structured, Unstructured, Semi-structured)

- Learning to examine different types of data available for the data curation process.
- Characteristics and examples of structured, unstructured and semi structured data.
- Learning to assess various data sensitivities and ensuring caution while dealing with sensitive, encrypted, masked data

- **Practical (5 Hours)**
  - Demonstration of features and functionalities of various open-source Data Curation Tools.
  - Hand on experience on popular open-source data curation tools.
  - Learning to examine the data to be curated and identifying any error or missing values in the data.

Data Collection & Acquisition Methods - 23 hours

- **Theory (13 Hours)**
  - Understanding the steps in Data Collection, Defining the goals of Data Collection & understanding what data to be collected
  - Learning about various methods of data collection- Surveys, Observational methods and Experimental Methods- Real-time data collection from sensors, IoT devices, APIs, and social media.
  - Learning to plan the approach- Primary & Secondary data sources
  - Learning about Data Acquisition techniques- Web scrapping, APIs and Data Feeds.
  - Learning about Ethical considerations in Data Collection- Informed consents, Data Privacy & confidentiality, Ethical guidelines and best practices.
  - Understanding Importance of data cleaning and learning about Common data quality issues.
  - Learning about Data Transformation techniques – Data Normalization, Data Standardization & Data Encoding.
  - Learning about automated data cleaning techniques using AI tools.
  - Learning about Data deduplication, outlier detection, and handling unstructured data.
  - Learning about various Data enrichment techniques- Augmenting with External Data Sources, text enrichment.
- **Practical (10 Hours)**
  - Understanding various real-world data collection scenarios.
  - Gaining practical experience with popular open-source data collection tools.
  - Hands-on practice sessions focusing on Data acquisition techniques, metadata management and dataset publishing
  - Learning to apply ethical considerations in data collection- informed consent, data privacy and confidentiality, and adherence to ethical guidelines and best practices.
  - Engaging in practical exercises to share and preserve research data through Data Tools.

  - Practical exercises using data cleaning and preprocessing tools- Talend Wrangler
  - Learning about handling missing data- techniques and Imputation methods

- Learning the practical exercises for Data Normalization & Data Standardization.
- Demonstration on automated data cleaning tools.
- Learning about Data enrichment techniques- Text Enrichment

Data Integration, Storage and Visualization (15 hours)

- **Theory (5 Hours)**
  - Understanding the concepts & architecture of Data Warehouse
  - Understanding the difference between data lakes and data warehouses
  - Overview of Database Management System
  - Understanding distributed databases and Cloud data governance
  - Introduction to cloud storage solutions- Security & scalability & compliance
  - Understanding various methods & standards to present the data.
  - Introduction to Open-source data visualization tools like Plotly and Matplotlib.
  - Learning about interactive dashboards and real-time reporting techniques for data visualization.
- **Practical (10 Hours)**
  - Learning about various Data Integration techniques- (Extract, Transform, Load) process, Data merging and consolidation
  - Practical exercises using data integration tools.
  - Learning to combine various data from multiple sources into a unified view, for comprehensive analysis and reporting.
  - Demonstration of popular cloud storage solutions
  - Hand on experience on Open-source data visualization tools like Plotly and Matplotlib.
  - Exercises on new templates and real time reporting dashboards for data visualization.

Data Quality and Governance (9 Hours)

- **Theory (4 Hours)**
  - Understanding key metrics for assessing data quality and learning techniques for measuring data quality.
  - Learning about techniques to ensure data accuracy and data consistency.
  - Learning about Data Lineage & data cataloging tools
  - Learning about Data governance framework and automation using AI tools.
- **Practical (5 Hours)**
  - Hand on experience on techniques for maintaining data accuracy and data consistency.
  - Hands on experience on implementing data cataloging tools.

Advanced Data Management Techniques (8 Hours)

- **Theory (3 Hours)**
  - Understanding Data governance frameworks along with relevant case studies of successful implementation to provide real world context.
  - Understanding about AI-assisted data curation tools and big data tools.
  - Understanding Cloud based data platforms and data management services.

- **Practical (5 Hours)**
    - Hands on experience on automated data curation AI tools on multiple tasks- tagging, quality assessments and data integration.
    - Learning to use big data tools and their applications in managing large datasets.

Application of Data Curation (12 Hours)

- Practical (12 Hours)
    - Working on real world Data curation problems sourced from Industry partners.
    - Learning to apply learned skills and knowledge in real world scenario
    - Learning to collaborate and work as a team for common objectives.
    - Presenting the project findings to the peers and instructor and learning to incorporate constructive feedback.

Key Learning Outcomes:

Duration: 45 hours	**Duration: 75 hours**
Theory	**Practical**
Participants will set up the Python environment, explore variables and data types, understand conditional statements (if, else, elif) and loops (for, while), and learn to define and call functions in Python.	Participants will install Python and an IDE, write and execute basic Python scripts, perform operations with variables and data types, use conditional statements and loops, and create functions within Python programs.
Participants will understand the basics of file handling in Python, explore Python lists and dictionaries, and learn about essential libraries like NumPy, Matplotlib, and Pandas.	Participants will practice reading from and writing to files, performing operations with lists and dictionaries, and using libraries like NumPy, Matplotlib, and Pandas. They will also develop and implement a project integrating the concepts learned.
Participants will be introduced to Artificial Intelligence (AI), covering its basic concepts, evolution, and key components such as Machine Learning, Deep Learning, Computer Vision, and Natural Language Processing (NLP). They will explore AI applications in various industries like healthcare, finance, education, and entertainment, and learn about Generative AI tools and use cases. Additionally, Participants will gain foundational knowledge in Data Science, including handling large datasets, data preprocessing techniques (e.g., cleaning, normalization), and statistical concepts like distributions, variance, and data sampling. Ethical considerations, including AI bias and privacy concerns, will also be addressed.	Participants will gain hands-on experience with popular AI tools, open-source Generative AI tools, and low-code/no-code AI platforms. They will explore AI use cases in industries such as healthcare, finance, education, and entertainment. Additionally, Participants will apply preprocessing techniques, such as data cleaning, normalization, and handling missing data, to real-world datasets.

Participants will define data curation and understand its scope, evaluating business needs and contexts for curating data. They will study case studies showcasing the impact of effective data curation and learn about different data types (structured, unstructured, and semi-structured), including their characteristics and examples. Additionally, Participants will understand how to assess data sensitivities and handle sensitive, encrypted, or masked data with caution.	Participants will get hands-on experience with open-source data curation tools, exploring their features and functionalities. They will examine data to be curated, identifying errors or missing values, and practice curating data using these tools.
Participants will understand the steps involved in data collection, define its goals, and identify the types of data to be collected. They will learn about various data collection methods, such as surveys, observational methods, and experimental methods, including real-time data collection from sensors, IoT devices, APIs, and social media. The module also covers the planning process for primary and secondary data sources, data acquisition techniques (web scraping, APIs, data feeds), and the ethical considerations in data collection, such as informed consent, data privacy, confidentiality, and adherence to ethical guidelines.	Participants will gain practical experience by exploring real-world data collection scenarios and using popular open-source data collection tools. Hands-on practice will focus on data acquisition techniques, metadata management, and dataset publishing. Additionally, Participants will apply ethical considerations in data collection, ensuring compliance with informed consent, privacy, and confidentiality practices. Practical exercises will also cover sharing and preserving research data using data tools.
This module covers the importance of data cleaning and addresses common data quality issues. Participants will explore data transformation techniques such as normalization, standardization, and encoding. It also includes an introduction to automated data cleaning using AI tools, along with data deduplication, outlier detection, and handling unstructured data. Furthermore, the theory will cover data enrichment techniques, including augmenting data with external sources and text enrichment.	Participants will engage in practical exercises using data cleaning and preprocessing tools. They will practice handling missing data with various techniques and imputation methods. The module includes hands-on activities for data normalization, data standardization, and demonstrating the use of automated data cleaning tools. Additionally, Participants will explore data enrichment techniques, focusing on text enrichment.
This module introduces the concepts and architecture of data warehouses, including the difference between data lakes and data warehouses. It covers an overview of database management systems (DBMS) and explores distributed databases and cloud data governance. The theory also includes an introduction to cloud storage solutions, focusing on aspects such as security, scalability, and compliance.	Participants will practice various data integration techniques, including the Extract, Transform, Load (ETL) process, data merging, and consolidation. Practical exercises will involve using data integration tools to combine data from multiple sources into a unified view for comprehensive analysis and reporting. The module will also include a demonstration of popular cloud storage solutions.
This module focuses on key metrics for assessing data quality and techniques for measuring it. It covers methods to ensure data	Participants will gain hands-on experience with techniques for maintaining data accuracy and consistency. They will also implement

accuracy and consistency, introduces data lineage and cataloging tools, and provides an understanding of data governance frameworks, including automation using AI tools.	data cataloging tools to manage and track data, ensuring effective governance and traceability.
This module introduces various methods and standards for presenting data. It covers open-source data visualization tools like Plotly and Matplotlib, along with interactive dashboards and real-time reporting techniques to visualize and communicate data insights effectively.	Participants will gain hands-on experience with open-source data visualization tools such as Plotly and Matplotlib. They will also engage in exercises to create new templates and develop real-time reporting dashboards to enhance their data visualization skills.
This module covers data governance frameworks, providing real-world context through case studies of successful implementations. It also introduces AI-assisted data curation tools, big data tools, and cloud-based data platforms and data management services.	Participants will gain hands-on experience with AI tools for automated data curation, including tasks such as tagging, quality assessments, and data integration. Additionally, they will explore the use of big data tools and their applications in managing large datasets.
Work with industry on real projects.	Participants will work on real-world data curation problems sourced from industry partners, applying their skills in practical scenarios. They will collaborate as a team to achieve common objectives and present their findings to peers and instructors, incorporating constructive feedback into their work.
This module focuses on exploring career paths in data curation, enhancing communication skills, developing problem-solving techniques, and fostering teamwork and collaboration.	Participants will engage in role-playing exercises and group activities designed to enhance teamwork, communication, and collaboration skills.
Classroom Aid	
Computers/Laptops, Projector and Screen, Smart Board, Chair, Computer Table	
Tools, Equipment and Other Requirements	
Installation of Python, including development environments like Jupyter Notebook, Anaconda Distribution, Comprehensive libraries such as pandas, numpy, matplotlib, scikit-learn (for Python)	

Annexure

Trainer Requirements

Trainer Prerequisites						
Minimum Educational Qualification	Specialization	Relevant Industry Experience		Training/Assessment Experience		Remarks
		Years	Specialization	Years	Specialization	
B. Tech in Computer Science/ IT/ allied branches Or MCA Or BCA Or M Sc (IT/CS) Or NIELIT A- Level Qualified		1	-	-	-	1 Years of experience in teaching

Trainer Certification	
Domain Certification	**Platform Certification**
Certified to TOT for the Standalone NOS: **NIE/ITS/N1016** "**Fundamentals of Data Curation using Python**" or equivalent as per NCrF. Minimum accepted score is 80%	Recommended that the Trainer is certified for the Job Role: "Trainer (VET and skills)", mapped to the Qualification Pack: "MEP/Q2601, v2.0" or equivalent as per NCrF. Minimum accepted score is 80%

Assessor Requirements

Assessor Prerequisites						
Minimum Educational Qualification	**Specialization**	**Relevant Industry Experience**		**Training/Assessment Experience**		**Remarks**
		Years	***Specialization***	***Years***	***Specialization***	
B. Tech in Computer Science/ IT/ allied branches Or		2	-	-	-	2 Years of Experience in teaching

M Sc (IT/CS) Or MCA Or BCA						

Assessor Certification	
Domain Certification	**Platform Certification**
Certified to TOA for the Standalone NOS: **NIE/ITS/N1016** "**Fundamentals of Data Curation using Python**" or equivalent as per NCrF. Minimum accepted score is 80%	Recommended that the assessor is certified for the Job Role: "Trainer (VET and skills)", mapped to the Qualification Pack: "MEP/Q2701, v2.0" or equivalent as per NCrF. Minimum accepted score is 80%

Assessment Strategy

This section includes the processes involved in identifying, gathering, and interpreting information to evaluate the Candidate on the required competencies of the program.

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as

per the defined learning outcomes and assessment criteria.

The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. competence acquired by the candidate can be obtained by conducting Theory (Online) examination.

About Examination Pattern:

1. The question papers for the theory exams are set by the Examination wing (assessor) of NIELIT HQS.

2. The assessor assigns roll number.

3. The assessor carries out theory online assessments. Theory examination would be conducted online and the paper comprise of MCQ

4. Pass percentage would be 50% marks.

5. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

References

Glossary

Term	Description
Declarative Knowledge	Declarative knowledge refers to facts, concepts and principles that need to be known and/or understood in order to accomplish a task or to solve a problem.
Key Learning Outcome	Key learning outcome is the statement of what a learner needs to know, understand and be able to do in order to achieve the terminal outcomes. A set of key learning outcomes will make up the training outcomes. Training outcome is specified in terms of knowledge, understanding (theory) and skills (practical application).
OJT (M)	On-the-job training (Mandatory); trainees are mandated to complete specified hours of training on site
OJT (R)	On-the-job training (Recommended); trainees are recommended the specified hours of training on site
Procedural Knowledge	Procedural knowledge addresses how to do something, or how to perform a task. It is the ability to work, or produce a tangible work output by applying cognitive, affective or psychomotor skills.
Training Outcome	Training outcome is a statement of what a learner will know, understand and be able to do upon the completion of the training.
Terminal Outcome	Terminal outcome is a statement of what a learner will know, understand and be able to do upon the completion of a module. A set of terminal outcomes help to achieve the training outcome.

Acronyms and Abbreviations

Term	Description
QP	Qualification Pack
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards